ADEN GUO

Calgary, AB · 587-888-4188 · aden.guowe@gmail.com · linkedin.com/in/aden-guo-b39743362

TECH STACK

- **Programming & Data Science:** Python, JavaScript, TypeScript, SQL, MATLAB, C++, NumPy, Pandas, SciPy, scikit-learn, PyTorch, OpenCV
- **GIS & Survey Tools:** ArcGIS, QGIS, Google Earth Engine, PostGIS, GeoPandas, GDAL/PROJ, RTK GNSS workflows, Total Stations, Digital leveling
- **Workflow & Version Control:** Git, GitHub, VS Code, Jupyter, Anaconda, virtualenv, Docker (basic)
- **Sensor Simulation & Estimation:** IMU/GNSS simulation, Extended Kalman Filter (EKF)
- **Proprietary & Survey Software:** AutoCAD, Civil 3D, MicroStation, Agisoft Metashape, Trimble Business Center (TBC)

EDUCATION

Bachelor of Science: Geomatics Engineering [Graduated]
University of Calgary - Calgary, AB
Relevant Skills:

- Photogrammetry, 3D modeling, and digital imaging workflows
- Geospatial data analysis, error modeling, and quality control
- Spatial database design, SQL, and geospatial data mining
- Project management, technical reporting, and collaborative problem-solving
- Applied geomatics workflow through capstone and ENGO 500 field survey camp

PROFESSIONAL EXPERIENCE

Engineering Capstone Project Lead - *University of Calgary*
Sep 2024 - Apr 2025

- Compared RTK GNSS, range finder, and odometer techniques to assess mapping accuracy and efficiency.
- Operated drones for photogrammetric capture and executed spatial intersection workflows for 3D modeling.
- Processed and validated geospatial datasets with photogrammetry tools and Python-based analysis pipelines.
- Authored and presented the final report at the SaGES Project Presentation.

Shift Lead / Team Trainer - *Gongcha*
Nov 2020 - May 2024

- Trained and mentored staff to improve consistency and teamwork.
- Balanced service demands and operational tasks under time constraints.
- Strengthened team coordination, problem-solving, and day-to-day execution.

PROJECTS

Photogrammetry Pipeline - Bundle Adjustment & Spatial Intersection

- Modular photogrammetry workflow using synthetic and real-world datasets for spatial intersection and bundle adjustment.
- Implemented camera models, pose estimation, and 3D reconstruction with C++ and Python.
- Built residual, optimization, and quality-assessment components to improve reconstruction accuracy.

ML-Driven Sensor Simulation + EKF Sensor Fusion

- Simulation framework for IMU/GNSS trajectory generation with machine learning and EKF fusion.
- Developing RNN/Transformer-based models for trajectory simulation.
- Integrating EKF for real-time estimation of pose, velocity, and orientation.
- Designing modular training, visualization, and evaluation pipelines.